

Predictions: Theory and Practice

Part II: Subjective judgement

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Evaluation

Intro

Goals

- learn how to test and compare prediction based on observed outcomes
- understand calibration diagrams
- comparing predictions (experts or models) with realized scores
- understand inference (confidence intervals and tests) for those tools

Setting

- we now observe a sample of
 - outcomes y
 - predictions $\hat{p}$ (or $\hat{y}$)
 - sometimes multiple predictions
 - and potentially some information x
- we want to test properties like
 - is a prediction calibrated
 - is a prediction better/worse than another?
 - is a prediction using information x or not

Recap: optimal predictions

- **sharpness**: Concentration/boldness of prediction
- **calibration**: Consistency of prediction with realized probabilities
- **accuracy**: expected score for proper scoring rules
 - tightly connected to optimal (expected utility maximizing) decision making
- **idealness** with respect to information x : using information x correctly
 - implies calibration
 - ideal prediction is the most accurate of all predictions that are based on info x
- we discussed those properties only for event predictions
 - here we will also look at real-valued targets

from theory to empirics

- all properties were discussed as theoretical properties of the random variables y and $\hat{p}/\hat{y}$
- now we have a sample $((y_1, \hat{p}_1), \dots, (y_n, \hat{p}_n))$
- to test a theoretical property like

$$E[y - \hat{p}] = 0$$

on a sample one approach is to substitute expectations ($E[\dots]$) by their empirical analogon, averages $\frac{1}{n} \sum_i [\dots]$

$$\frac{1}{n} \sum_i (y_i - \hat{p}_i)$$

Calibration

Calibration

- calibration of $\hat{p}$ for y :

$$E[y|\hat{p}] = \hat{p}$$

- can be judged by looking at calibration curve $E[y|\hat{p}]$
- for a set of predictions and target outcomes
 - $(y_1, \hat{p}_1), \dots, (y_n, \hat{p}_n)$

we can compute averages of y for each value of $\hat{p}$

$$\text{calibration curve}(c) = \frac{1}{n_c} \sum_{\hat{p}_i=c} y_i$$

where n_c denotes the number of predictions with $\hat{p}_i = c$.

Example

- Let's look at two examples
- Example A:
 - $\hat{p}$: Doctors assigning probability of pneumonia
 - y : radiology finds pneumonia
 - $n = 1531$
- Example B:
 - $\hat{p}$: US weather forecast prediction rain
 - y : actual rain that day
 - $n > 100k$ (based on 87 different weather stations)

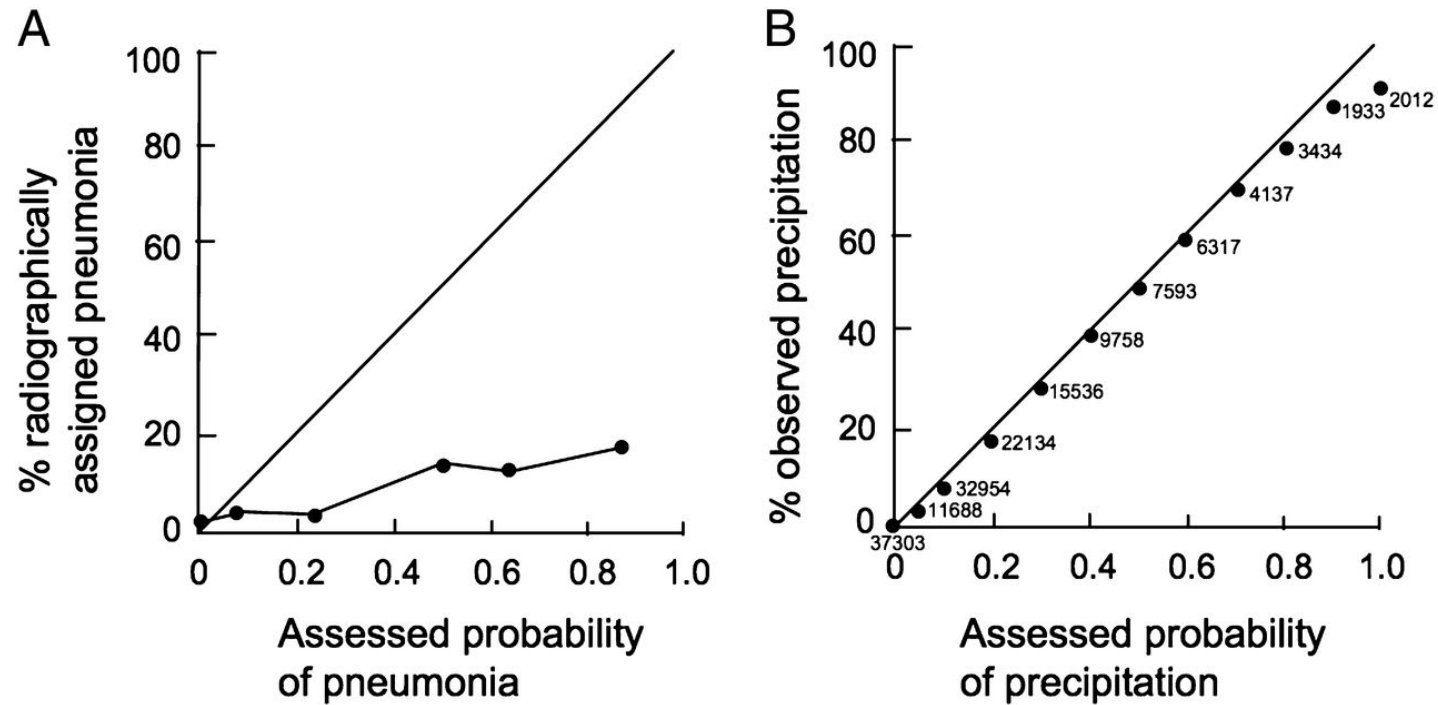
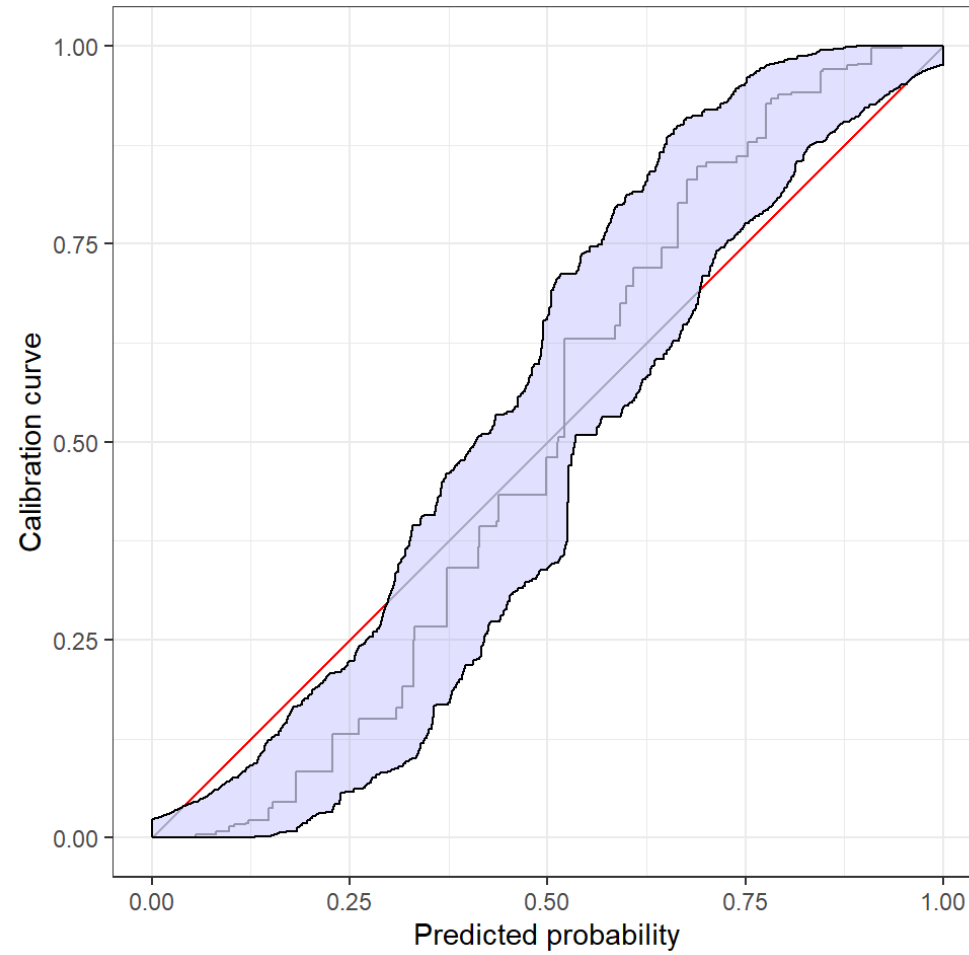


Illustration of two extremes in expert calibration. (A) Assessment of probability of pneumonia (based on observed symptoms) in 1,531 first-time patients (B) Once-daily US Weather Service precipitation forecasts for 87 stations are compared with actual occurrence of precipitation. The small numbers adjacent to each point report the number of forecasts. (Morgan 2014)

Inference

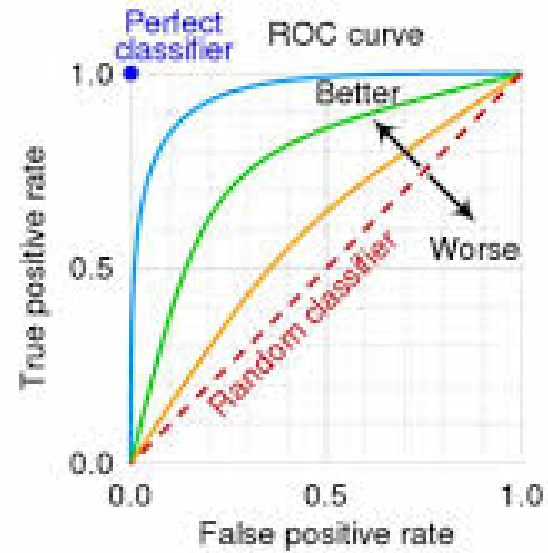
- If we observe a certain calibration curve, we want to know how much uncertainty remains
- we will not cover the statistical details
- confidence intervals for $E[y|\hat{p} = c]$
 - beware of multiple testing!
- Tests for $H_0 : E[y|\hat{p} = c] = 0$
 - p-value: likelihood of observing deviation from zero if prediction calibrated

Simulated example with confidence intervals



ROC (sidenote)

- ROC curves
 - for event prediction (“classification”)
 - can be used for any signal (does not need to be a probability)



Some roc curves. Each curve corresponds to a prediction model

Calibration for reals

- Now the case with a real-valued target $y \in \mathbb{R}$
- A probabilistic prediction is now a distribution $\hat{P}$
 - could be defined by a cdf $\hat{F}_{\hat{P}}(x) = \hat{P}(Y \leq x)$
 - or a parametric distribution (e.g., a Gaussian distribution $P = \mathcal{N}(0, 1)$)
- for simplicity we assume a continuous distribution (a smooth cdf without “jumps”)

PIT

- one approach is to use the Probability Integral Transform (PIT)
 - defined by the predicted probability that the target is lower than actually observed

$$PIT(y, \hat{P}) = F_{\hat{P}}(y)$$

calibrated implies uniform PIT

If a probabilistic prediction for a real-valued target y is calibrated, i.e.,

$$Y|\hat{P} = \hat{P},$$

then the PIT is uniformly distributed, i.e.,

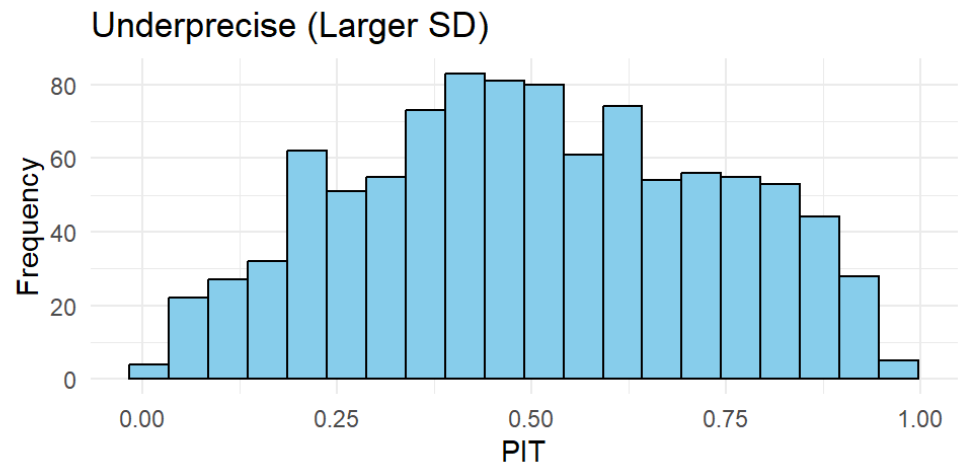
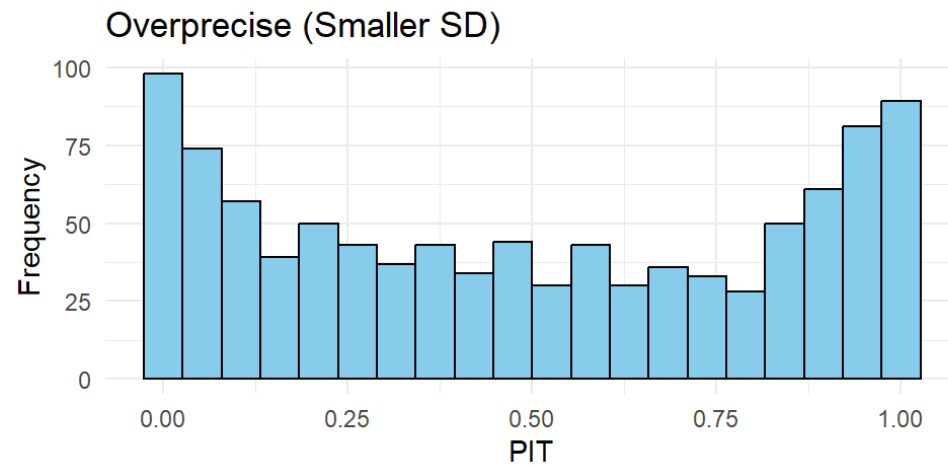
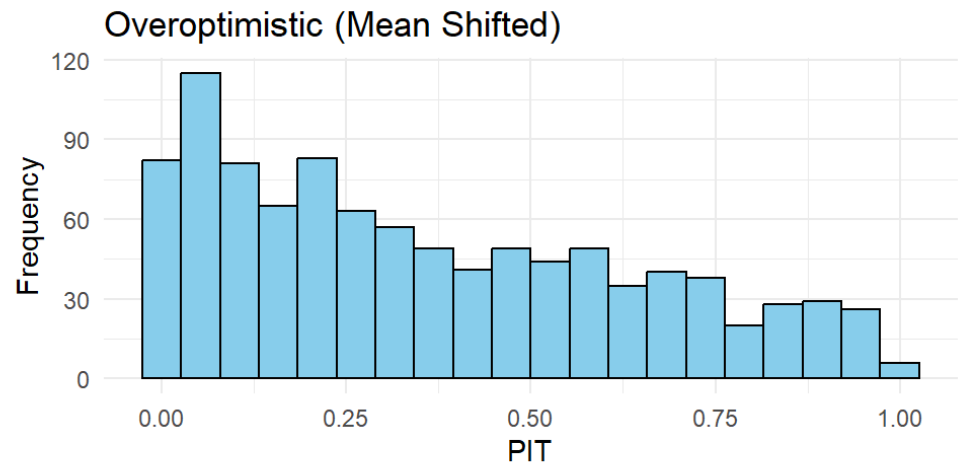
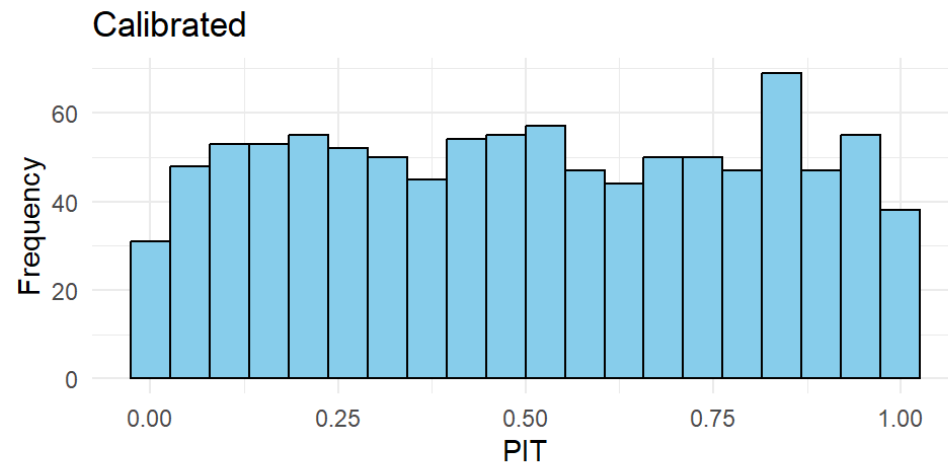
$$PIT(y, \hat{P}) \sim U[0, 1].$$

PIT histograms

- for a sample of predictions with real-valued targets we can
 - compute a sample of PIT values
 - plot a histogram of them
 - compare the histogram to a uniform distribution

simulation

- Let us look at a simulated pit histogram
- we have $y \sim \mathcal{N}(0, 1)$
- and predictions
 - calibrated $\hat{P} = \mathcal{N}(0, 1)$
 - overoptimistic $\hat{P} = \mathcal{N}(\textcolor{red}{0.5}, 1)$
 - overprecise $\hat{P} = \mathcal{N}(0, \textcolor{red}{.7})$
 - underprecise $\hat{P} = \mathcal{N}(0, \textcolor{red}{1.5})$
- $n = 1000$



Calibration of prediction intervals

- probability predictions $\hat{F}$ can be used to construct prediction intervals

$$[\hat{F}^{-1}(\alpha/2), \hat{F}^{-1}(1 - \alpha/2)]$$

- a prediction interval $[\hat{l}, \hat{u}]$ at level $1 - \alpha$ should contain the true outcome with probability $1 - \alpha$:

$$P(y \in [\hat{l}, \hat{u}]) = 1 - \alpha$$

- **empirical coverage**: proportion of observations where the outcome falls inside the interval

$$\widehat{\text{coverage}} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{y_i \in [\hat{l}_i, \hat{u}_i]\}$$

- deviations from the nominal level signal miscalibration:
 - **undercoverage** ($\widehat{\text{coverage}} < 1 - \alpha$): intervals too narrow — prediction is overprecise
 - **overcoverage** ($\widehat{\text{coverage}} > 1 - \alpha$): intervals too wide — prediction is underprecise

Accuracy

scoring rules

- we showed that the expected score is a measure of accuracy
 - if scoring rule is proper
- given a sample of predictions and target observations we can
 - compute the score $s_i = s(y_i, \hat{p}_i)$
 - compute the average score $\frac{1}{n} \sum_i s_i$ to approximate $S(\hat{p})$
- if we compare two predictions $\hat{p}_A$ and $\hat{p}_B$, the more accurate prediction will obtain a higher average score (in the long run)

inference

- comparing the expected value / mean of two populations is a standard problem in statistics
- common approaches are
 - compute the two average scores with confidence intervals
 - compute the average score difference $\Delta s = s(y_i, \hat{p}_A) - s(y_i, \hat{p}_B)$
 - test an equal means null hypothesis, $H_0 : E[s(y_i, \hat{p}_A)] = E[s(y_i, \hat{p}_B)]$
- often a prediction is compared to the **ignorant prediction** with $\hat{p} = 0.5$

Example: AI forecast bench

- forecastbench is a benchmark for prediction ability of large language models (<https://www.forecastbench.org/>)
- idea:
 - if models understand world, they can predict it
 - benchmark is not cheatable (no training spill) as realization in future

Example: results

Rank	Org	Model	Overall
1		Superforecaster median forecast	70.6
2		green tree	68.1
3		Grok 4.20 (Preview)	67.7
4		Cassi ensemble_2_crowdadj	67.6
4		yellow mouse	67.6
6		GPT-5-2025-08-07 (zero shot with crowd forecast)	67.1
7		Gemini-3-Pro-Preview (zero shot with crowd forecast)	67.0
7		Foresight-32B	67.0
9		Grok-4-0709 (zero shot with crowd forecast)	66.8
9		Claude-Opus-4-5-20251101 (zero shot with crowd forecast)	66.8

Sidenote on the score used above:

- ranking based on the quadratic score (Brier score): $BS = \frac{1}{n} \sum_i (y_i - \hat{p}_i)^2$
- FRI introduced a new index using the **square root** for better interpretability:

$$BI = (1 - \sqrt{BS}) \times 100$$

- $BI = 70$ means $\sqrt{BS} = 0.30$, i.e. the same RMSE as a forecaster who always predicts $\hat{p} = 0.70$ when $y = 1$ and $\hat{p} = 0.30$ when $y = 0$
- **Problem:** per observation, the Brier Index equals:

$$BI_i = 1 - \sqrt{(y_i - \hat{p}_i)^2} = 1 - |y_i - \hat{p}_i|$$

which is the **linear score** and therefore **not proper** (see script1.2).

For details see their [blogpost](#)

Accuracy for probabilistic predictions

- For real-valued targets $y \in \mathbb{R}$, proper scoring rules also exist
- **Log score** (also called ignorance score):

$$S_{\log}(y, \hat{P}) = \log \hat{f}(y)$$

where $\hat{f}$ is the density of the predicted distribution $\hat{P}$

- **CRPS** (Continuous Ranked Probability Score):

$$\text{CRPS}(y, \hat{F}) = \int_{-\infty}^{\infty} \left(\hat{F}(x) - \mathbf{1}\{y \leq x\} \right)^2 dx$$

- Both are **proper scoring rules** for probabilistic predictions of real-valued targets
 - higher expected score = more accurate prediction

CRPS: two representations

- The CRPS has two useful equivalent representations
- **Via quadratic (Brier) scores of events** $\{Y \leq x\}$ for all thresholds x :

$$\text{CRPS}(y, \hat{F}) = \int_{-\infty}^{\infty} \underbrace{\left(\hat{F}(x) - \mathbf{1}\{y \leq x\} \right)^2}_{\text{Brier score for event } \{Y \leq x\}} dx$$

- **Via quantile scores** integrated over all levels $\tau \in [0, 1]$:

$$\text{CRPS}(y, \hat{F}) = 2 \int_0^1 \underbrace{(\mathbf{1}\{y \leq \hat{q}_\tau\} - \tau) (\hat{q}_\tau - y)}_{\text{quantile score at level } \tau} d\tau$$

where $\hat{q}_\tau = \hat{F}^{-1}(\tau)$ is the τ -quantile of $\hat{P}$

- The CRPS can be seen as an aggregate accuracy measure across all events / all quantile levels

accuracy for point predictions

- For real-valued targets probability predictions are often too complicated
- instead, we use point predictions and evaluate their accuracy via loss functions $l(y, \hat{y})$
 - e.g. the quadratic loss $l(y, \hat{y}) = (y - \hat{y})^2$
 - loss functions are “scoring rules for point predictions”
- here we focus on mean predictions (other point predictions will be covered in Part III)
 - we want to predict the mean / expected value of P

consistent loss functions

Definition: consistent loss function

We say a loss function $l(y, \hat{y})$ is **consistent for a mean prediction** if the mean of the true distribution minimizes the expected loss, i.e.,

$$\arg \min_{\hat{y}} E_{y \sim P} [l(y, \hat{y})] = E_{y \sim P} [y]$$

- the quadratic loss function is **consistent for a mean prediction**
 - result is derived in the same way as properness of qsr
- there also exists other consistent loss functions for the mean

target properties and loss functions

- for other target properties (e.g., the median, quantiles) we need other loss functions
- the property targeted by a point prediction (e.g., the mean) and the loss function should always match
- **not** to do:
 - estimate a mean prediction and measure accuracy with absolute loss $l(\hat{y}, y) = |y - \hat{y}|$

calibration of point predictions

- for real-valued targets y with a point prediction $\hat{y}$
- Mincer-Zarnowitz regression

$$E[y - \hat{y} | x] = \beta_0 + \beta_1 \hat{y}$$

- if a point prediction is calibrated, i.e.

$$E[y | \hat{y}] = \hat{y},$$

then $\beta_0 = 0$ and $\beta_1 = 0$.

- can be tested with statistical tools from regression analysis like the F -test
- MZ-regressions test a form of calibration for point predictions
 - “mean-calibration”

Pitfalls

Extreme events

- In the evaluation of predictions, we should not condition on extreme events y
 - Some forecasters are famous for predicting important/extreme events (financial crises, Covid pandemic,...)
 - looking at accuracy only for extreme events makes extreme predictions look better
- Instead, condition on extreme forecasts x
 - such comparisons are theoretically valid (Lerch et al. 2017)

Sample size

- small sample size and large correlation
- inference on calibration and accuracy depends on sample size and correlation
- confidence intervals are large and tests have low power if
 - sample size is small
 - observations are correlated (time series, realizations depend on same underlying circumstances)

Heterogeneity

- heterogeneity in predictive properties
 - e.g., a weather prediction model is better under specific atmospheric conditions
 - e.g., a stock market experts is better in low-inflation environment
- we focused on testing properties on the entire sample
- predictions may be calibrated only on subset
- predictions may be more or less accurate under specific situations

Not covered

- some common tools for evaluating predictions were not covered here
- calibration tests for point predictions
 - e.g., backtesting Value-at-Risk predictions

Summary

prediction type	calibration	accuracy
event probability	calibration curve	log score, qsr
prob. distribution	PIT histogram	log score, CRPS
point prediction	MZ-regression	loss functions

Summary

- for **event** predictions we can
 - test calibration with calibration curves
 - compare predictive accuracy with average scores
- for **real-valued** targets we can
 - test calibration of probabilistic predictions with PIT histograms
 - compare predictive accuracy of probabilistic predictions with the log score or CRPS
- for **point predictions** of real-valued targets
 - we compare predictive accuracy with loss functions
 - loss functions should be consistent for the property under consideration
 - the quadratic error loss is consistent for mean predictions
- inference and sampling uncertainty is crucial

References

- Lerch, Sebastian, Thordis L Thorarinsdottir, Francesco Ravazzolo, and Tilmann Gneiting. 2017. “Forecaster’s Dilemma: Extreme Events and Forecast Evaluation.” *Statistical Science*, 106–27.
- Morgan, M Granger. 2014. “Use (and Abuse) of Expert Elicitation in Support of Decision Making for Public Policy.” *Proceedings of the National Academy of Sciences* 111 (20): 7176–84.